

2. Solve the following system of equations using Gaussian Elimination:

i) $x + y + z = 2$, $x + 2y + 3z = 5$, $2x + 3y + 4z = 11$

Solution : $[A:b] = \begin{bmatrix} 1 & 1 & 1 & : & 2 \\ 1 & 2 & 3 & : & 5 \\ 2 & 3 & 4 & : & 11 \end{bmatrix} \xrightarrow[R_3 - 2R_1]{R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 & : & 2 \\ 0 & 1 & 2 & : & 3 \\ 0 & 1 & 2 & : & 7 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 1 & : & 2 \\ 0 & 1 & 2 & : & 3 \\ 0 & 1 & 2 & : & 7 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 1 & 1 & : & 2 \\ 0 & 1 & 2 & : & 3 \\ 0 & 0 & 0 & : & 4 \end{bmatrix}$$

Here $\rho(A) = 2$; $\rho(A:b) = 3$. $\therefore$ No solution.

ii) $x - y + z = 8$, $2x + 3y - z = -2$, $3x - 2y - 9z = 9$

Solution : $[A:b] = \begin{bmatrix} 1 & -1 & 1 & : & 8 \\ 2 & 3 & -1 & : & -2 \\ 3 & -2 & -9 & : & 9 \end{bmatrix} \xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & -1 & 1 & : & 8 \\ 0 & 5 & -3 & : & -18 \\ 0 & 1 & -12 & : & -15 \end{bmatrix}$

$$\begin{bmatrix} 1 & -1 & 1 & : & 8 \\ 0 & 5 & -3 & : & -18 \\ 0 & 1 & -12 & : & -15 \end{bmatrix} \xrightarrow{R_3 - 1/5 R_2} \begin{bmatrix} 1 & -1 & 1 & : & 8 \\ 0 & 5 & -3 & : & -18 \\ 0 & 0 & -57/5 & : & -57/5 \end{bmatrix}$$

$$\frac{-12}{1} + \frac{3}{5} = \frac{-60+3}{5}$$

$$\frac{-15}{1} + \frac{18}{5} = \frac{-75+18}{5} = \frac{-57}{5}$$

$$\begin{aligned} x - y + z &= 8 \\ x + 3 + 1 &= 8 \\ \boxed{x &= 4} \end{aligned}$$

unique solution

$$\boxed{z = 1}$$

$$5y - 3z = -18$$

$$5y - 3 = -18$$

$$5y = -15$$

$$\boxed{y = -3}$$

(2)

$$2 \text{ iii) } 3x + y - 6z = -10; 2x + y - 5z = -8$$

$$6x - 3y + 3z = 0$$

$$\left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 2 & 1 & -5 & -8 \\ 6 & -3 & 3 & 0 \end{array} \right] \xrightarrow{\substack{R_2 - \frac{2}{3}R_1 \\ R_3 - 2R_1}} \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 0 & 1/3 & -1 & -4/3 \\ 0 & -5 & 15 & 20 \end{array} \right]$$

$$1 - \frac{2}{3}; -5 + \frac{12}{3}; -8 + \frac{20}{3}$$

$$-5 - x\left(\frac{1}{3}\right) = 0$$

$$-5 = x/3$$

$$x = -15$$

$$\left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 0 & 1/3 & -1 & -4/3 \\ 0 & -5 & 15 & 20 \end{array} \right] \xrightarrow{R_3 + 15R_2} \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 0 & 1/3 & -1 & -4/3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$20 + 15\left(-\frac{4}{3}\right) = 0$$

$\rho(A) = \rho(A:b) = 2 < 3$. The system of equations have infinite no. of solutions

$$y/3 - z = -\frac{4}{3}$$

$$\text{let } z = k$$

$$\frac{y}{3} - k = -\frac{4}{3}$$

$$y/3 = -\frac{4}{3} + k$$

$$\frac{y}{3} = \frac{3k - 4}{3}$$

$$y = 3k - 4$$

$$3x + y = -10$$

$$3x + (3k - 4) = -10$$

$$3x + 3k - 4 = -10$$

$$3x + 3k = -6 \Rightarrow 3x = -6 - 3k$$

$$x = -2 - k$$

4) For the below system of equations :

(3)

$$x + y + z = 1$$

$$x + 3z = -2 + s$$

$$x - y + rz = 3$$

a) For what value of r & s is this system of linear equation is inconsistent?

b) For what values of r & s does this system of linear equations have infinitely many solutions?

c) For what values of r & s does this system of linear equation have a unique solution?

Solution:-

$$[A:b] = \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 1 & 0 & 3 & : & s-2 \\ 1 & -1 & r & : & 3 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & -1 & 2 & : & s-3 \\ 0 & -2 & r-1 & : & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & -1 & 2 & : & s-3 \\ 0 & -2 & r-1 & : & 2 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & -1 & 2 & : & s-3 \\ 0 & 0 & r-5 & : & 8-2s \end{bmatrix}$$

$$2 - 2s + 6 \Rightarrow 8 - 2s$$

a) For $r=5$ and $s \neq 0$, systems of equations are inconsistent

b) For $r=5$ & $s=4$, Infinite Solution

c) For $r \neq 5$, unique solution

(4)

3) For which numbers a does elimination break down

1) permanently 2) temporarily?

$$ax + 3y = -3$$

$$4x + 6y = -6$$

Solve for x & y after fixing temporary breakdown by a row exchange.

Solution

$$\begin{bmatrix} a & 3 & : & -3 \\ 4 & 6 & : & -6 \end{bmatrix}$$

If $a=2$, elimination will fail

If $a=0$, elimination will stop after a row exchange

If $a=0$, we have $\begin{bmatrix} 0 & 3 & : & -3 \\ 4 & 6 & : & -6 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 4 & 6 & : & -6 \\ 0 & 3 & : & -3 \end{bmatrix}$

$$3y = -3 \quad 4x + 6y = -6$$

$$\Rightarrow \boxed{y = -1} \quad 4x + 6(-1) = -6$$

$$4x = 0$$

$$\boxed{x = 0}$$

$$\therefore (x, y) = (0, -1)$$

5) What three elimination matrices E_{21} , E_{31} & E_{32} put A into upper triangular form $E_{32} \cdot E_{31} \cdot E_{21} \cdot A = U$? Using these, compute the matrix L and U to factor $A = LU$?

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 4 & 0 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -6 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -6 \end{bmatrix}$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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7) Apply elimination to produce the factors L & U for $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & 2 \\ 3 & 1 & 3 \end{bmatrix}$. Is $A = LU$ possible? Explain.

Write down the permutation matrices if any. Also decompose $PA = LDU$?

Solution $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & 2 \\ 3 & 1 & 3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 3 & 1 & 3 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 3 & 1 & 3 \end{bmatrix} \xrightarrow{R_3 - 3R_1} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 0 & -5 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 0 & -5 & -3 \end{bmatrix} \xrightarrow{R_3 + 5R_2} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 17 \end{bmatrix}$$

U

Let us verify whether $A = LU$

$$LU = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -5 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 17 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 3 & 1 & 3 \end{bmatrix}$$

Since $LU \neq A$, this is not factorizable

∴ $PA = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & 2 \\ 3 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 3 & 1 & 3 \end{bmatrix}$

∴ $PA = LU$

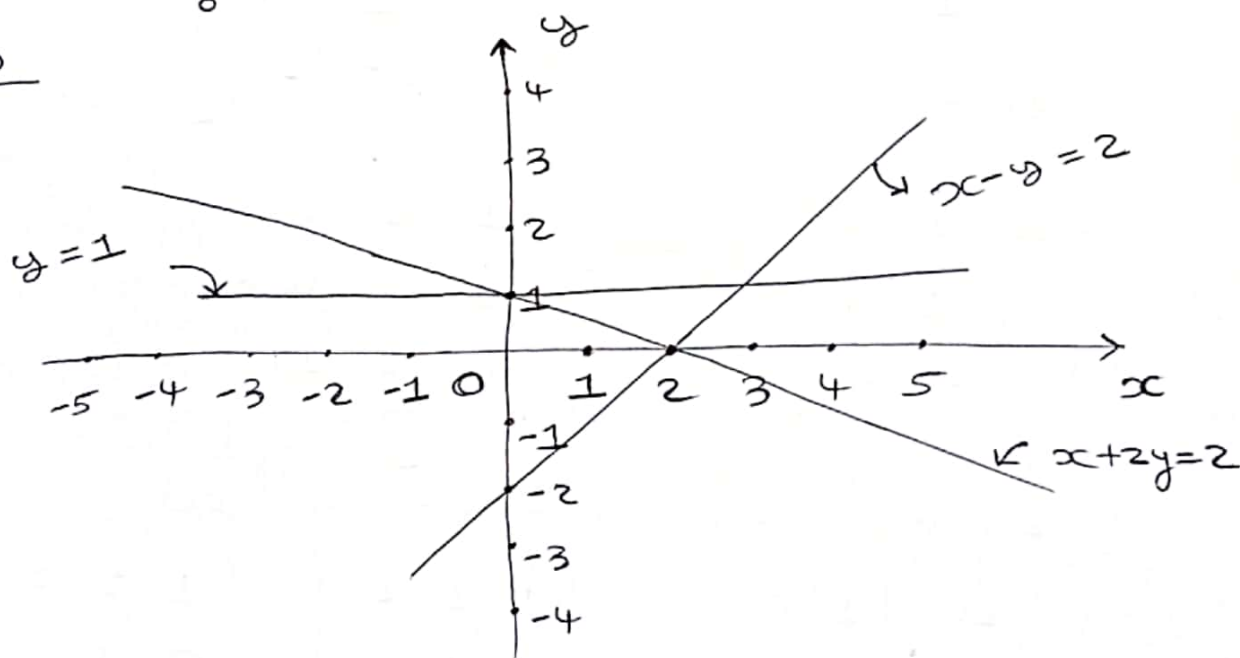
⑥

$$LDU = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 3 & -5 & 1 \end{bmatrix}}_L \cdot \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 17 \end{bmatrix}}_D \cdot \underbrace{\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}}_U$$

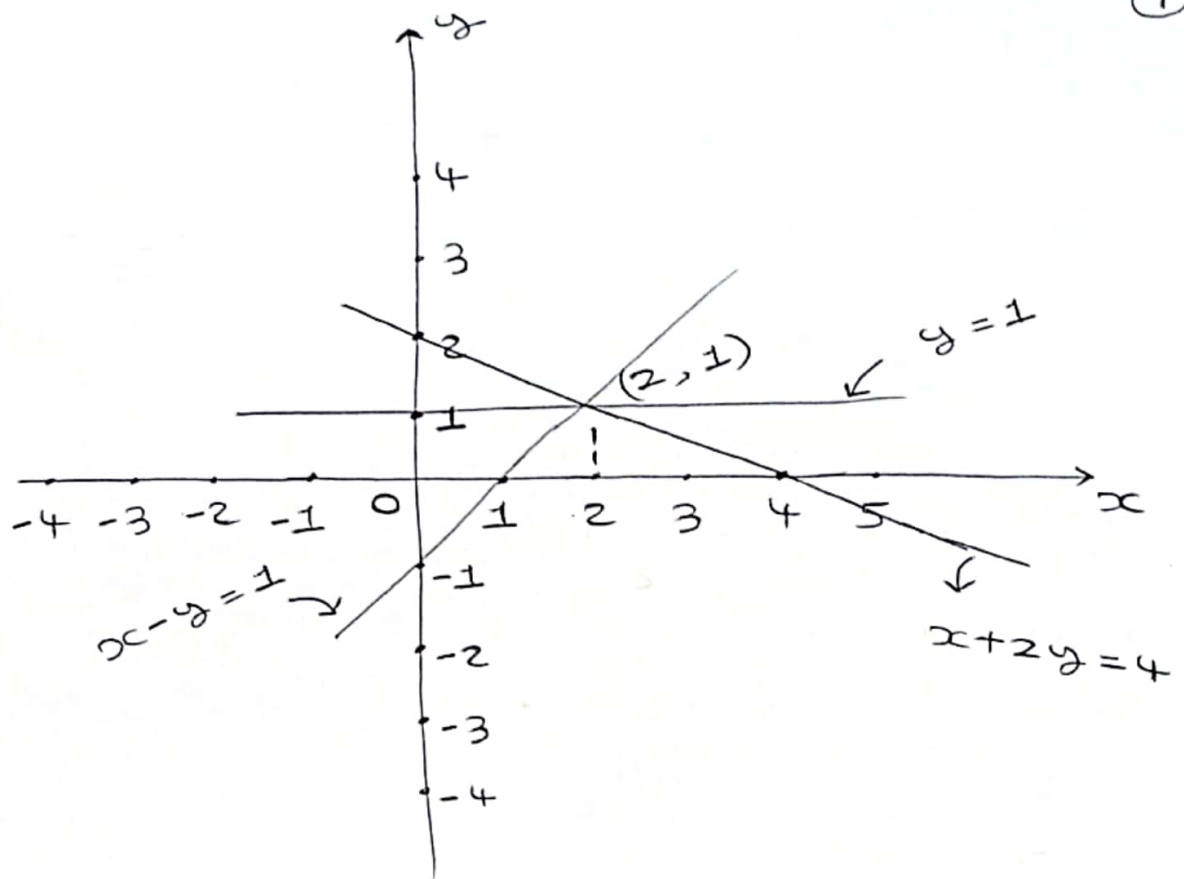
$$\boxed{PA = LDU}$$

① sketch these three lines and decide if the equations are solvable $\circ x+2y=2, x-y=2, y=1$.
 What happens if all the right-hand side are zero?
 Is there any non-zero choice of right hand sides that allow the three lines to intersect at the same point?

Solution



These equations are not solvable as they do not have a common point of intersection. If all RHS are zero, we get a trivial solution $x=0$ & $y=0$. If $x+2y=4, x-y=1$ & $y=1$; we get a common point of intersection $(2,1)$.



12) Joseph invests a total of 10,000\$ in three accounts, one paying 5% interest, another paying 8%, and the third paying 9% interest. The annual interest earned on the three investments last year was 770\$. The amount invested at 9% was twice the amount invested at 5%. How much was invested at each rate?

Solution:- Let x be the amount invested at 5% rate. Let y be the amount invested at 8% rate & Let z be the amount invested at 9%.

The annual interest earned = 770\$

$$x + y + z = 10,000$$

$$0.05x + 0.08y + 0.09z = 770$$

$$2x = z \Rightarrow 2x - z = 0$$

⑧

$$\begin{bmatrix} 1 & 1 & 1 & : & 10,000 \\ 0.05 & 0.08 & 0.09 & : & 770 \\ 2 & 0 & -1 & : & 0 \end{bmatrix} \xrightarrow[\substack{R_3 - 2R_1 \\ R_2 - 0.05R_1}]{R_3 - 2R_1} \begin{bmatrix} 1 & 1 & 1 & : & 10,000 \\ 0 & 0.03 & 0.04 & : & 270 \\ 0 & -2 & -3 & : & -20,000 \end{bmatrix}$$

$$-2 - x(0.03) = 0$$

$$-2 = x(0.03)$$

$$\boxed{x = \frac{-2}{0.03}}$$

$$\begin{bmatrix} 1 & 1 & 1 & : & 10,000 \\ 0 & 0.03 & 0.04 & : & 270 \\ 0 & -2 & -3 & : & -20,000 \end{bmatrix} \xrightarrow{R_3 + \frac{2}{0.03}R_2} \begin{bmatrix} 1 & 1 & 1 & : & 10,000 \\ 0 & 0.03 & 0.04 & : & 270 \\ 0 & 0 & -\frac{1}{3} & : & -2,000 \end{bmatrix}$$

$$-3 + \frac{2}{0.03} \times 0.04 = -3 + \frac{8}{3} = -\frac{1}{3}$$

$$-20,000 + \frac{2}{0.03} \times 270 = -2,000$$

$$-\frac{z}{3} = -2000 \Rightarrow \boxed{z = 6000}$$

$$0.03y + 0.04z = 270$$

$$0.03y + 0.04(6000) = 270$$

$$0.03y = 30$$

$$\boxed{y = 1000}$$

$$x + y + z = 10000$$

$$x + 1000 + 6000 = 10,000$$

$$\boxed{x = 3,000}$$

11) Determine the equation of the polynomial of degree two whose graph passes through the points $(1, 6)$, $(2, 6)$ and $(3, 2)$. Find y when $x = -2$?

Solution :- Equation of the polynomial of degree 2 is given by :- $y = ax^2 + bx + c$

$$\text{For } (1, 6) \Rightarrow a + b + c = 6$$

$$\begin{aligned} \text{For } (2, 6) \Rightarrow a(4) + b(2) + c &= 6 \\ 4a + 2b + c &= 6 \end{aligned}$$

$$\begin{aligned} \text{For } (3, 2) \Rightarrow a(3^2) + b(3) + c &= 2 \\ 9a + 3b + c &= 2 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 4 & 2 & 1 & 6 \\ 9 & 3 & 1 & 2 \end{array} \right] \xrightarrow{\begin{array}{l} R_2 - 4R_1 \\ R_3 - 9R_1 \end{array}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & -3 & -18 \\ 0 & -6 & -8 & -52 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & -3 & -18 \\ 0 & -6 & -8 & -52 \end{array} \right] \xrightarrow{R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & -3 & -18 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$\begin{array}{lll} \boxed{c=2} & -2b - 3c = -18 & a + b + c = 6 \\ & -2b - 6 = -18 & a + 6 + 2 = 6 \\ & -2b = -12 & \boxed{a = -2} \\ & \boxed{b = 6} & \end{array}$$

$$\therefore y = -2x^2 + 6x + 2$$

when $x = -2$:-

$$y = -2(4) + 6(-2) + 2$$

$$y = -8 - 12 + 2$$

$$\boxed{y = -18}$$

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13) By using Gaussian elimination method, balance the chemical reaction equation:



Solution $\text{C}_2\text{H}_6 \rightarrow \begin{bmatrix} 2 \\ 6 \\ 0 \end{bmatrix}$; $\text{O}_2 \rightarrow \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$; $\text{H}_2\text{O} \rightarrow \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$

$$\text{CO}_2 \rightarrow \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$x_1 \begin{bmatrix} 2 \\ 6 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = x_3 \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$x_1 \begin{bmatrix} 2 \\ 6 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 & -1 & : & 0 \\ 6 & 0 & -2 & 0 & : & 0 \\ 0 & 2 & -1 & -2 & : & 0 \end{bmatrix} \xrightarrow{R_2 - 3R_1} \begin{bmatrix} 2 & 0 & 0 & -1 & : & 0 \\ 0 & 0 & -2 & 3 & : & 0 \\ 0 & 2 & -1 & -2 & : & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 & -1 & : & 0 \\ 0 & 0 & -2 & 3 & : & 0 \\ 0 & 2 & -1 & -2 & : & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 2 & 0 & 0 & -1 & : & 0 \\ 0 & 2 & -1 & -2 & : & 0 \\ 0 & 0 & -2 & 3 & : & 0 \end{bmatrix}$$

$$-2x_3 + 3x_4 = 0$$

$$2x_2 - x_3 - x_4 = 0$$

$$2x_1 - x_4 = 0$$

Let $x_4 = K$

$$2x_1 = K$$

$$x_1 = K/2$$

$$2x_3 = 3x_4$$

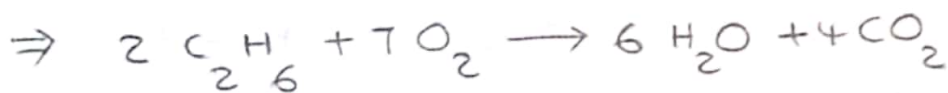
$$\boxed{x_3 = \frac{3}{2}K}$$

$$2x_2 - \frac{3}{2}K - 2K = 0$$

$$2x_2 = \frac{7}{2}K$$

$$\boxed{x_2 = \frac{7}{4}K}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} K/2 \\ 7/4 K \\ 3/2 K \\ K \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 6 \\ 4 \end{bmatrix}$$



10) compute the symmetric LDL^T factorization of $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$

Solution

$$b - x \cdot a = 0$$

$$b = x \cdot a$$

$$x = b/a$$

$$\begin{bmatrix} a & b \\ b & d \end{bmatrix} \xrightarrow{R_2 - b/a R_1} \begin{bmatrix} a & b \\ 0 & d - \frac{b^2}{a} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ b/a & 1 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & d - \frac{b^2}{a} \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix}$$

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6) compute LU and LDU factorization for the Symmetric matrix ? Find the four conditions on the entries of A to get $A=LU$ with four pivots.

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$$

Solution

$$\begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix} \xrightarrow[\begin{smallmatrix} R_3 - R_1 \\ R_4 - R_1 \end{smallmatrix}]{R_2 - R_1} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix}$$

$$\begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & b-a & c-a & d-a \end{bmatrix}$$

$$\begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & b-a & c-a & d-a \end{bmatrix} \xrightarrow{R_4 - R_2} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix}$$

$$\begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix} \xrightarrow{R_4 - R_3} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$$

The condition to get four pivots are :-
 $a \neq 0$; $b \neq a$; $c \neq b$ & $d \neq b$.

8) Decompose A by using cholesky's Method :-

$$A = \begin{bmatrix} 65 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix}$$

solution $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$

$$l_{11} = \sqrt{a_{11}} = \sqrt{65} = 8.062$$

$$l_{21} = \frac{a_{21}}{l_{11}} = \frac{15}{8.062} = 1.8605$$

$$l_{22} = \sqrt{a_{22} - l_{21}^2} = \sqrt{55 - 1.8605^2} = 7.179$$

$$l_{31} = \frac{a_{31}}{l_{11}} = \frac{55}{8.062} = 6.822$$

$$l_{32} = \frac{a_{32} - l_{31} \cdot l_{21}}{l_{22}} = \frac{225 - \{6.822 \times 1.8605\}}{7.179}$$

$$\boxed{l_{32} = 29.57}$$

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$$l_{33} = \sqrt{a_{33} - l_{31}^2 - l_{32}^2}$$

$$l_{33} = \sqrt{9.79 - 6.822^2 - 29.57^2}$$

$$\boxed{l_{33} = 7.62}$$

$$\therefore L = \begin{bmatrix} 8.062 & 0 & 0 \\ 1.8605 & 7.179 & 0 \\ 6.822 & 29.57 & 7.62 \end{bmatrix}$$

$$\therefore \boxed{A = L \cdot L^T}$$

$$\Rightarrow A = \begin{bmatrix} 8.062 & 0 & 0 \\ 1.8605 & 7.179 & 0 \\ 6.822 & 29.57 & 7.62 \end{bmatrix} \cdot \begin{bmatrix} 8.062 & 1.8605 & 6.822 \\ 0 & 7.179 & 29.57 \\ 0 & 0 & 7.62 \end{bmatrix}$$

9) compute the inverse of the following matrices by Gauss-Jordan method :-

$$\text{a) } \begin{bmatrix} 5 & 7 & 9 \\ 4 & 3 & 8 \\ 7 & 5 & 6 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$$

Solution :- a) $\begin{bmatrix} 5 & 7 & 9 \\ 4 & 3 & 8 \\ 7 & 5 & 6 \end{bmatrix}$

$$[A : I] = \left[\begin{array}{ccc|ccc} 5 & 7 & 9 & 1 & 0 & 0 \\ 4 & 3 & 8 & 0 & 1 & 0 \\ 7 & 5 & 6 & 0 & 0 & 1 \end{array} \right] \xrightarrow[R_3 - \frac{1}{5}R_1]{R_2 - \frac{4}{5}R_1} \left[\begin{array}{ccc|ccc} 5 & 7 & 9 & 1 & 0 & 0 \\ 0 & -\frac{13}{5} & \frac{4}{5} & -\frac{4}{5} & 1 & 0 \\ 0 & -\frac{24}{5} & -\frac{33}{5} & -\frac{1}{5} & 0 & 1 \end{array} \right]$$

$$\begin{aligned} 4 - x \cdot 5 &= 0 & 7 - y \cdot 5 &= 0 & 3 - \frac{28}{5} & ; 8 - \frac{36}{5} \\ 4 &= x \cdot 5 & 7 &= 5 \cdot y & 5 - \frac{49}{5} & ; 6 - \frac{63}{5} \\ x &= \frac{4}{5} & y &= \frac{7}{5} & & \end{aligned}$$

$$\begin{bmatrix} 5 & 7 & 9 & : & 1 & 0 & 0 \\ 0 & -\frac{13}{5} & \frac{4}{5} & : & -\frac{4}{5} & 1 & 0 \\ 0 & -\frac{24}{5} & -\frac{33}{5} & : & -\frac{7}{5} & 0 & 1 \end{bmatrix} \xrightarrow{R_3 - \frac{24}{13} R_2} \begin{bmatrix} 5 & 7 & 9 & : & 1 & 0 & 0 \\ 0 & -\frac{13}{5} & \frac{4}{5} & : & -\frac{4}{5} & 1 & 0 \\ 0 & 0 & -\frac{525}{65} & : & \frac{1}{13} & -\frac{24}{13} & 1 \end{bmatrix}$$

$$-\frac{24}{5} - x \left\{ -\frac{13}{5} \right\} = 0$$

$$-\frac{24}{5} + \frac{13x}{5} = 0$$

$$\frac{-24}{5} = \frac{-13x}{5}$$

$$x = \frac{24}{13}$$

$$0 \quad -\frac{24}{5} - \frac{24}{13} \left\{ -\frac{13}{5} \right\} = 0$$

$$\frac{-7}{5} - \frac{24}{13} \left(-\frac{4}{5} \right) = \frac{-7}{5} + \frac{96}{65}$$

$$= \frac{-91+96}{65}$$

$$\frac{-33}{5} - \frac{24}{13} \cdot \frac{4}{5}$$

$$= \frac{-33}{5} - \frac{96}{65}$$

$$= \frac{-429-96}{65}$$

$$= \frac{-525}{65}$$

$$\frac{-33 \times 13}{99}$$

$$\frac{33 \times}{-429}$$

$$-429$$

$$-96$$

$$-525$$

$$\begin{bmatrix} 5 & 7 & 9 & : & 1 & 0 & 0 \\ 0 & -\frac{13}{5} & \frac{4}{5} & : & -\frac{4}{5} & 1 & 0 \\ 0 & 0 & -\frac{525}{65} & : & \frac{1}{13} & -\frac{24}{13} & 1 \end{bmatrix} \xrightarrow{R_1 + \frac{585}{525} R_3} \begin{bmatrix} 5 & 7 & 0 & : & 1 & 0 & 0 \\ 0 & -\frac{13}{5} & \frac{4}{5} & : & -\frac{4}{5} & 1 & 0 \\ 0 & 0 & -\frac{525}{65} & : & \frac{1}{13} & -\frac{24}{13} & 1 \end{bmatrix}$$

$$9 + x \left(\frac{-525}{65} \right) = 0$$

$$9 = \frac{525x}{65}$$

$$x = \frac{585}{525}$$

$$9 + \frac{585}{525} \times \frac{-525}{65} = 0$$

$$1 + \frac{585}{525} = \frac{1,110}{525}$$

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$$\begin{bmatrix} 5 & 7 & 0 & : & 2.11 & 0 & 0 \\ 0 & -2.6 & 0.8 & : & -0.8 & 1 & 0 \\ 0 & 0 & -8.07 & : & 0.07 & -1.84 & 1 \end{bmatrix} \longrightarrow$$

$$0.8 + x(-8.07) = 0$$

$$0.8 = x(8.07)$$

$$\boxed{x = 0.099}$$

Assignment - 01 - Questions

3) Investigate the values of λ & μ such that

$$x + 3y + 5z = 9$$

$$x - y + 2z = 1$$

$$2x + 2y + \lambda z = \mu$$

has i) unique solution ii) infinitely many solution
iii) no solution :-

Solution

$$\begin{bmatrix} 1 & 3 & 5 & : & 9 \\ 1 & -1 & 2 & : & 1 \\ 2 & 2 & \lambda & : & \mu \end{bmatrix} \xrightarrow[R_3 - 2R_1]{R_2 - R_1} \begin{bmatrix} 1 & 3 & 5 & : & 9 \\ 0 & -4 & -3 & : & -8 \\ 0 & -4 & \lambda - 10 & : & \mu - 18 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 5 & : & 9 \\ 0 & -4 & -3 & : & -8 \\ 0 & -4 & \lambda - 10 & : & \mu - 18 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 3 & 5 & : & 9 \\ 0 & -4 & -3 & : & -8 \\ 0 & 0 & \lambda - 7 & : & \mu - 10 \end{bmatrix}$$

i) for $\lambda \neq 7$ & $\mu \neq 10 \Rightarrow \rho(A) = \rho(A:b) = n$; unique solution

ii) for $\lambda = 7$ & $\mu = 10 \Rightarrow \rho(A) = \rho(A:b) = 2 < n$; Infinitely many solution

iii) for $\lambda = 7$ & $\mu \neq 10$ (OR) $\lambda \neq 7$ & $\mu = 10$; $\rho(A) \neq \rho(A:b)$
∴ NO solution

2) Determine the equation of the polynomial $y=f(x)$ of degree 2 whose graph passes through points $(1,6)$, $(2,3)$ and $(3,2)$

Solution :-

$$y = ax^2 + bx + c$$

$$6 = a(1) + b(1) + c$$

$$\Rightarrow a + b + c = 6 \quad \text{--- (1)}$$

$$a(4) + b(2) + c = 3$$

$$4a + 2b + c = 3 \quad \text{--- (2)}$$

$$a(9) + b(3) + c = 2$$

$$9a + 3b + c = 2 \quad \text{--- (3)}$$

$$\therefore \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 4 & 2 & 1 & : & 3 \\ 9 & 3 & 1 & : & 2 \end{bmatrix} \xrightarrow[R_3 - 9R_1]{R_2 - 4R_1} \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & -2 & -3 & : & -21 \\ 0 & -6 & -8 & : & -52 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & -2 & -3 & : & -21 \\ 0 & -6 & -8 & : & -52 \end{bmatrix} \xrightarrow{R_3 - 3R_2} \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & -2 & -3 & : & -21 \\ 0 & 0 & 1 & : & 11 \end{bmatrix}$$

$$\boxed{c = 11}$$

$$-2b - 3c = -21$$

$$-2b - 33 = -21$$

$$-2b = 12$$

$$\boxed{b = -6}$$

$$a + b + c = 6$$

$$a - 6 + 11 = 6$$

$$a + 5 = 6$$

$$\boxed{a = 1}$$

$$\therefore \boxed{y = x^2 - 6x + 11}$$

4) Solve the system of equation by using LU decomposition :-

$$x + 2y + 3z = 14 ; 2x + 3y + 4z = 20 ; 3x + 4y + z = 14$$

$$\text{Solution :- } \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 2 & 3 & 4 & : & 20 \\ 3 & 4 & 1 & : & 14 \end{bmatrix} \xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 0 & -1 & -2 & : & -8 \\ 0 & -2 & -8 & : & -28 \end{bmatrix}$$

(18)

$$\begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 0 & -1 & -2 & : & -8 \\ 0 & -2 & -8 & : & -28 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 0 & -1 & -2 & : & -8 \\ 0 & 0 & -4 & : & -12 \end{bmatrix}$$

$$-4z = -12$$

$$z = 3$$

$$-y - 2z = -8$$

$$-y - 6 = -8$$

$$y = 2$$

$$x + 2(2) + 3(3) = 14$$

$$x + 4 + 9 = 14$$

$$x = 1$$

1) Determine the solution of system of equations:

$$2x_1 + x_2 - x_3 + 2x_4 = 5$$

$$4x_1 + 5x_2 - 3x_3 + 6x_4 = 9$$

$$-2x_1 + 5x_2 - 2x_3 + 6x_4 = 4$$

$$4x_1 + 11x_2 - 4x_3 + 8x_4 = 2$$

Solution :-

$$\begin{bmatrix} 2 & 1 & -1 & 2 & : & 5 \\ 4 & 5 & -3 & 6 & : & 9 \\ -2 & 5 & -2 & 6 & : & 4 \\ 4 & 11 & -4 & 8 & : & 2 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 - 2R_1 \\ R_3 + R_1 \\ R_4 - 2R_1 \end{matrix}} \begin{bmatrix} 2 & 1 & -1 & 2 & : & 5 \\ 0 & 3 & -1 & 2 & : & -1 \\ 0 & 6 & -3 & 8 & : & 9 \\ 0 & 9 & -2 & 4 & : & -8 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & -1 & 2 & : & 5 \\ 0 & 3 & -1 & 2 & : & -1 \\ 0 & 6 & -3 & 8 & : & 9 \\ 0 & 9 & -2 & 4 & : & -8 \end{bmatrix} \xrightarrow{\begin{matrix} R_3 - 2R_2 \\ R_4 - 3R_2 \end{matrix}} \begin{bmatrix} 2 & 1 & -1 & 2 & : & 5 \\ 0 & 3 & -1 & 2 & : & -1 \\ 0 & 0 & -1 & 4 & : & 11 \\ 0 & 0 & 1 & -2 & : & -5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & -1 & 2 & : & 5 \\ 0 & 3 & -1 & 2 & : & -1 \\ 0 & 0 & -1 & 4 & : & 11 \\ 0 & 0 & 1 & -2 & : & -5 \end{bmatrix} \xrightarrow{R_4 + R_3} \begin{bmatrix} 2 & 1 & -1 & 2 & : & 5 \\ 0 & 3 & -1 & 2 & : & -1 \\ 0 & 0 & -1 & 4 & : & 11 \\ 0 & 0 & 0 & 2 & : & 6 \end{bmatrix}$$

$$2x_4 = 6 \Rightarrow x_4 = 3$$

$$-x_3 + 4x_4 = 11$$

$$-x_3 + 12 = 11$$

$$-x_3 = -1$$

$$x_3 = 1$$

$$3x_2 - x_3 + 2x_4 = -1$$

$$3x_2 - 1 + 2(3) = -1$$

$$3x_2 - 1 + 6 = -1$$

$$3x_2 = -1 + 1 - 6$$

$$x_2 = -2$$

$$2x_1 + x_2 - x_3 + 2x_4 = 5$$

$$2x_1 - 2 - 1 + 2(3) = 5$$

$$2x_1 - 3 + 6 = 5$$

$$2x_1 + 3 = 5$$

$$2x_1 = 2$$

$$x_1 = 1$$

∴

$$x_1 = 1, x_2 = -2, x_3 = 1, x_4 = 3$$